FOR BEGINNERS, FROM ZERO

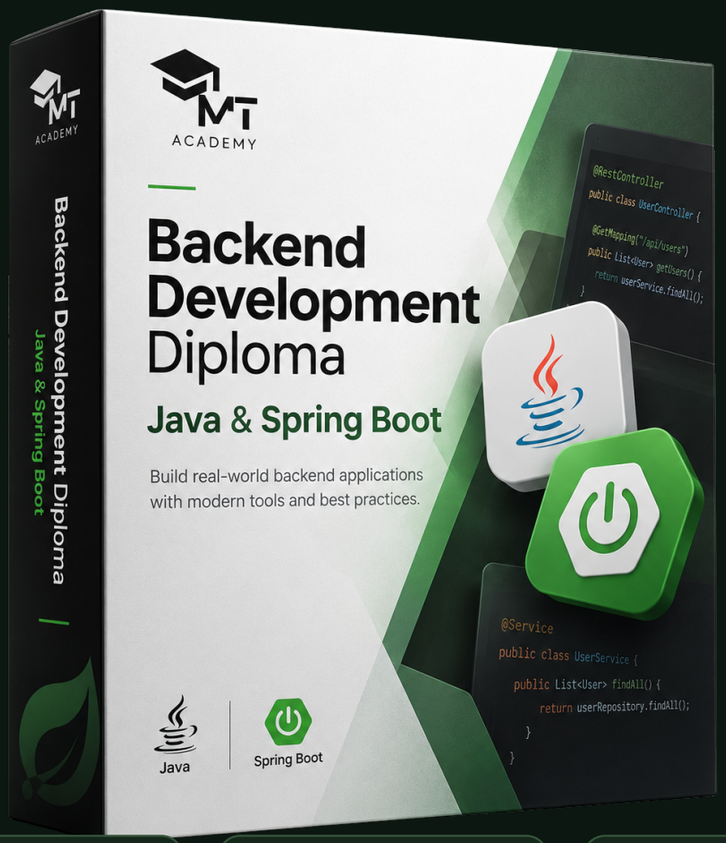
Backend Development Diploma

Java & Spring Boot

Overview & Roadmap

From your first Java program to a backend you can build, test, deploy and explain.

Mohamed Tamer
Trainer & Mentor



9	72	9
Training months	Live sessions	Practical projects

A clear commitment. A practical learning path.

9	36	72	360	9
Training months	Training weeks	Live sessions	Scheduled hours	Monthly projects

For beginners, from zero

Start with basic computer and file-management skills. Java 17 is the programming baseline; Spring Boot 4 and Spring 7 are introduced after Java and SQL.

Two live sessions each week

Each five-hour block includes explanation, live coding, student work, review and breaks. Across 72 sessions: 324 learning hours + 36 break hours = 360 scheduled hours.

Practice between sessions

Plan for 12-16 hours of independent practice each week. Complete assignments, finish project increments and apply review feedback.

Mentor and code review

Use the agreed support channels for guidance. Weekly submissions, rotating code reviews and monthly project feedback support individual progress. Track questions and apply the agreed revisions.

Java programming

01 Java 17 Fundamentals & Advanced Java

Getting started

JDK 17, IntelliJ IDEA, JShell, project setup, syntax, variables, primitive types, casting, operators and console input.

Programming & OOP

Conditions, switch expressions, loops, methods, overloading, classes, constructors, inheritance, encapsulation, composition and polymorphism.

Data structures & modern Java

Arrays, lists, sets, maps, enums, generics, Comparable / Comparator, nested classes, lambdas, method references, Streams and Optional.

Java 17 features & data handling

Records, sealed classes, instanceof pattern matching, text blocks, immutability, final, defensive copying, BigDecimal, Date / Time and regex.

Files, concurrency & networking

Exceptions, try-with-resources, I/O, NIO, binary files, serialization, threads, locks, executors, concurrent collections, sockets and HttpClient.

Debugging & practical foundations

Breakpoints, watches, stepping, JUnit basics and coding challenges. JDBC / JPA foundations follow SQL later in the path; asynchronous HTTP and a guided WebSocket example.

Teamwork & problem solving

02
VERSION CONTROL

Git, GitHub & Team Collaboration

Git basics: repositories, commits, history, diff and .gitignore.

Branches, merging, merge conflicts, remotes and GitHub workflows.

Pull requests, code review, team collaboration and useful project READMEs.

03
WORKING AS A TEAM

Agile, Scrum & Kanban Fundamentals

Agile values, iterative delivery and feedback.

Scrum roles, Product Backlog, Sprint Planning, Daily Scrum, Review and Retrospective.

Kanban boards, work in progress limits and tracking delivery.

04
UNDERSTANDING THE PRODUCT

Requirements Engineering & SRS

Stakeholders, requirement gathering and functional / non-functional requirements.

User stories, use cases, acceptance criteria and prioritization.

SRS structure, requirement validation, traceability and change management.

05
PROBLEM SOLVING

Data Structures, Algorithms & Problem Solving

Big O, arrays, linked lists, stacks, queues, hash tables and trees.

Recursion, searching, sorting and common problem-solving patterns.

Coding exercises: edge cases, explaining solutions and time / space trade-offs.

Code design & databases

06

CODE DESIGN

SOLID Principles & Design Patterns

SOLID: SRP, OCP, LSP, ISP and DIP, with Java refactoring practice.

Builder, Factory Method, Singleton, Adapter, Decorator, Facade and Proxy.

Chain of Responsibility, Observer, Strategy and Template Method; UML, trade-offs and pitfalls.

07

RELATIONAL DATABASES

SQL & MySQL Database Design

Schemas, tables, data types, CRUD, constraints, primary / foreign keys and relationships.

Joins, subqueries, aggregates, GROUP BY, HAVING, string / date functions and window functions.

Views, SQL modes, triggers and schema changes; JDBC, PreparedStatement, transactions and stored procedures.

08

DATABASE PERFORMANCE

MySQL Indexing & Query Optimization

How indexes work: primary, secondary, composite and covering indexes.

EXPLAIN, query plans, selectivity and finding expensive queries.

Improve filters, joins, sorting and pagination; compare results and index write costs.

Spring applications & security

09

BUILDING WEB APPLICATIONS

Spring Boot, REST APIs & Hibernate

Maven, Spring Boot 4, Spring 7, IoC, Dependency Injection, bean scopes and lifecycle.

HTTP, JSON, REST CRUD, validation, exception handling, PATCH and OpenAPI / Swagger.

Hibernate / JPA CRUD, layered APIs, MVC, Thymeleaf forms, MVC security, AOP and Actuator.

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PERSISTENCE & SCHEMA CHANGES

Spring Data JPA & Database Migrations with Flyway

Repositories, derived queries, JPQL, Criteria API, pagination and sorting.

Entity relationships, cascades, lazy / eager loading, N+1 queries, JOIN FETCH and transactions.

Flyway versioned migrations, schema history, repeatable migrations and database evolution.

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AUTOMATED TESTING

Unit Testing with JUnit, Mockito & MockMvc

JUnit assertions, test lifecycle, parameterized tests, exceptions, timeouts and TDD.

Mockito stubbing, verification, exception scenarios, Spring bean mocks and ReflectionTestUtils.

MockMvc tests for MVC / REST, service and repository checks, test profiles, SQL fixtures and coverage.

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APPLICATION SECURITY

Spring Security, JWT & OAuth 2.0

Servlets, Filters, SecurityFilterChain, users, PasswordEncoder and AuthenticationProvider.

Sessions, SecurityContext, roles / authorities, method security, CORS, CSRF and custom filters.

JWT, OAuth 2.0, OpenID Connect, PKCE, refresh tokens, social login, Keycloak and Spring Authorization Server.

Linux, Docker & Kubernetes

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WORKING IN LINUX

Linux & Command Line Fundamentals

Terminal basics, paths, files, directories and the Linux filesystem.

Permissions, users, processes, environment variables and command-line tools.

Pipes, redirects, searching files and logs; running and troubleshooting Java applications.

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DISTRIBUTED APPS & CONTAINERS

Microservices, Docker & Kubernetes

Microservice boundaries, DTOs, cloud-native configuration, Config Server, Eureka and API Gateway.

Resilience, RabbitMQ, Kafka, Spring Cloud Stream, metrics, logs, traces and OAuth 2.0 security.

Docker from scratch: images, containers, Dockerfile, Buildpacks, Jib, networks, volumes and Compose.

Kubernetes from scratch: Pods, Deployments, Services, ConfigMaps, Secrets, probes and rollouts; guided Helm / cloud deployment.

Docker and Kubernetes start with the basics, then move into guided deployments of your own services.

Integration testing & CI/CD

15 TESTING REAL INTEGRATIONS

Integration Testing with Testcontainers

Integration tests with real services running in disposable containers.

Testcontainers setup, MySQL containers, connection properties and database lifecycle.

Verify repositories, API flows and Flyway migrations; isolate test data and run tests in CI.

16 AUTOMATING DELIVERY

CI/CD with GitHub Actions

GitHub Actions workflows, events, jobs, runners and steps.

Build Java applications, run unit / integration tests and publish test reports.

Secrets, artifacts, container image builds and a guided deployment workflow.

Use automated checks to verify a change before you package or deploy the application.

Architecture & system design

17

SYSTEM DESIGN FOUNDATIONS

Software Architecture & System Design Fundamentals

Functional requirements, quality attributes, architecture boundaries and design trade-offs.

Capacity estimates, latency, throughput, availability and scalability.

Load balancing, caching, replication, partitioning / sharding and messaging fundamentals.

Consistency choices, failure handling and choosing a monolith or microservices.

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DESIGNING & EXPLAINING SYSTEMS

System Design Case Studies & Interview Preparation

Work through practical cases such as URL shortening, notifications and an order platform.

Define APIs, data models, request flows and service boundaries.

Identify bottlenecks, discuss scaling options and compare design alternatives.

Practise design interviews: clarify requirements, draw the system and explain trade-offs.

Start with the requirements. Explain the design, compare alternatives and test the assumptions.

AI tools & career preparation

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AI-ASSISTED CODING

AI-Assisted Development with Codex

Codex workflows: provide repository context, clear tasks and acceptance criteria.

Plan changes, implement small features, investigate bugs and generate useful tests.

Inspect diffs, run checks and review results; protect secrets and explain accepted code.

20

AI-ASSISTED PROJECT WORK

AI-Assisted Development with Claude Code

Claude Code setup, project context, focused prompts and practical coding workflows.

Explore a codebase, debug a failure, refactor safely and improve tests.

Review changes, manage permissions and verify outcomes before accepting a result.

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CAREER PREPARATION

CV, LinkedIn & Job Interview Preparation

Developer CV, LinkedIn profile, GitHub portfolio and project READMEs.

Job search strategy, relevant applications and clear project walkthroughs.

Coding, technical and behavioural interview practice; communicate your own contribution.

Your first five monthly projects

One project each training month, with assignments, live coding, a demo and review feedback.

01 MONTH 01

Expense Tracker Console

Build a console app with input validation, calculations, menus and clear control flow.

02 MONTH 02

Library & Lending Console

Model books, members and loans using OOP, collections and reusable operations.

03 MONTH 03

File Processing & Reporting Tool

Read and validate files, transform records and produce useful reports with clean Java design.

04 MONTH 04

Social Analytics Database & JDBC App

Design a relational schema, write analytical SQL and connect a Java application through JDBC.

05 MONTH 05

Employee Operations Portal

Build an employee CRUD application with REST endpoints, Spring MVC and persistent data.

From tested services to your capstone

One project each training month, with assignments, live coding, a demo and review feedback.

06	MONTH 06	**Tested Appointment Service**	Develop an appointment backend with JPA relationships, migrations and automated tests.
07	MONTH 07	**Secure Learning API**	Secure a learning application with authentication, authorization, token handling and security checks.
08	MONTH 08	**Banking Microservices Platform**	Connect banking services using containers, configuration, routing, messaging and observability.
09	MONTH 09	**Commerce Platform Capstone**	Combine requirements, APIs, persistence, security, tests, CI/CD and guided deployment in a final team project.

Build, explain, test and improve.

36 training weeks / 72 live sessions / 360 scheduled hours: 324 learning hours + 36 break hours.

Live coding with student practice

Follow a short implementation, then modify it yourself. Predict results, run the code, investigate a failure and explain the change.

Assignments with visible evidence

Submit code, test cases or queries after each session. Keep changes in Git and include clear instructions for running your work.

Monthly project checkpoints

Each month ends with a distinct project submission or demo. Review notes become concrete revision tasks before the next stage.

Team collaboration

Practise branches, pull requests, requirements, acceptance criteria and reviews. The capstone includes an individual contribution record.

Reviewed AI assistance

Codex and Claude Code are introduced after programming and testing foundations. Provide context, inspect diffs and verify every accepted change.

Five-hour session structure

25 min recap + 60 min concepts + 70 min guided coding + 75 min student practice + 25 min review + 15 min exit task. Add three 10-minute breaks: 300 minutes total.

Show the work. Explain the decisions.

A working backend portfolio

Submit nine monthly projects at increasing levels of complexity. Develop selected projects into stronger portfolio examples with clear requirements and run instructions.

Engineering evidence

Show validation, database integrity, security rules, automated tests, Git history, pull requests and deployment evidence where relevant.

An individual change task

Implement a small feature or bug fix, adjust a test and explain the result. Include an unaided task alongside reviewed AI-assisted work.

A defensible design

Explain the API, data model, dependency boundaries and an important trade-off. Respond to a changed requirement without relying on memorized diagrams.

Career preparation

Prepare a targeted CV, LinkedIn profile, project README and a concise project walkthrough. Practise coding, design and behavioural interview questions.

Clear levels of depth

Core Java, SQL, APIs, testing and security receive repeated practice. Kubernetes deployment is guided; sharding, service mesh and mTLS are introductory. Record personal gaps and a next-step plan.

First cohort: February 2027, proposed start.

Live teaching is in Arabic, with standard English technical terminology. The first cohort is limited to 15-20 students. The exact start day and weekly timetable are confirmed before the cohort begins.

EGYPTIAN STUDENTS

EGP 16,500

Full payment for the complete diploma

Installment total: EGP 18,000

Deposit at booking: EGP 4,000

7 monthly installments of EGP 2,000

INTERNATIONAL STUDENTS

USD 650

Full payment for the complete diploma

Installment total: USD 750

Deposit at booking: USD 150

6 monthly installments of USD 100

The deposit is part of the total, not an extra fee. Monthly installments start when teaching begins. The stated totals cover the complete diploma; payment dates move with any agreed change to the cohort start. Booking and withdrawal terms are provided before payment.